Can we automate tuning an EKF?

- "Tuning" a KF is the process of selecting $\Sigma^+_{\tilde{x},0}$, $\Sigma_{\widetilde{w}}$, and $\Sigma_{\tilde{v}}$.
- For linear systems, perhaps we can find $\Sigma_{\widetilde{w}}$ and $\Sigma_{\tilde{v}}$ from sensor datasheets, or from independent measurements.
- But, $\Sigma_{\widetilde{w}}$ and $\Sigma_{\tilde{v}}$ also describe, to an extent, inaccuracies in the state and measurement equations, and so are not trivial to determine even in the linear case.
- For nonlinear systems, the overall impact of w_k and v_k can be additionally difficult to quantify due to the nonlinearities of the state and measurement equations.
- It would be very nice if we could automate a way to estimate $\Sigma_{\widetilde{w}}$ and $\Sigma_{\tilde{v}}$ from data.

Literature search

- The earliest work that I'm aware of on this topic is a paper by Mehra.[1] This paper is still heavily cited.
  - The derivation is surprisingly complicated.
- Different variations of Mehra's approach appear in the literature, because of making different assumptions in different steps.
- Two of the best references I have found for putting the best practices of all of these into a single document are by Fraser.[2]

[1] Mehra, Raman. "On the identification of variances and adaptive Kalman filtering." *IEEE Transactions on automatic control* 15(2), 1970, 175–184.

[2] Cory T. Fraser and Steve Ulrich. "Adaptive extended Kalman filtering strategies for spacecraft formation relative navigation." *Acta Astronautica,* 178, 2021, 700–721.

Cory T. Faser, *Adaptive Extended Kalman Filtering Strategies for Autonomous Relative Navigation of Formation Flying Spacecraft*, MSAE Thesis, Carleton University, 2018.

Basic idea

- I'm not going to present the full derivation here. If you are interested, please confer documents cited on the prior slide.
  - But, I would like to give a "sketch" of the idea.
- We treat $\Sigma_{\widetilde{w}}$ and $\Sigma_{\tilde{v}}$ as unknown parameters of the system model. We can define this set of parameters as $\Theta = \{\Sigma_{\widetilde{w}}, \Sigma_{\tilde{v}}\}$.
- We wish to find the $\Sigma_{\widetilde{w}}$, $\Sigma_{\tilde{v}}$ that maximize the likelihood of Θ given the set of measurements $\mathbb{Z}_k$:

$$\Theta^* = \arg\max_{\Theta} \{L(\Theta \mid \mathbb{Z}_k)\}.$$

- To do this, we assume that Θ has a Gaussian distribution and substitute the Gaussian pdf in the computation of likelihood, and proceed from there.

Not a proof: Motivating form of $\boldsymbol{\Sigma}_{\widetilde{w}}$

- If we assume w_k to be additive, then the state equation says:

$$x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}.$$

- Using EKF as a basis, consider Steps 1a and 2b:

$$\hat{x}_k^- = \mathbb{E}[x_k \mid \mathbb{Z}_{k-1}] \approx f(\hat{x}_{k-1}^+, u_{k-1}) + \bar{w}_{k-1}$$
$$\hat{x}_k^+ = \mathbb{E}[x_k \mid \mathbb{Z}_k] \approx \hat{x}_k^- + L_k \underbrace{(z_k - \hat{z}_k)}_{\mu_k}.$$

- If we assume $\hat{x}_k^+ \approx x_k$, then $\hat{x}_k^+ - \hat{x}_k^- \approx w_k - \bar{w}_k = \widetilde{w}_{k-1}$.
  - But, we also see that $\hat{x}_k^+ - \hat{x}_k^- = L_k \mu_k$ (where μ_k is the innovation).
  - So, we can approximate: $\widetilde{w}_{k-1} \approx \hat{x}_k^+ - \hat{x}_k^- = L_k \mu_k$.
- The covariance that we seek:

$$\Sigma_{\widetilde{w}} = \mathbb{E}[\widetilde{w}_{k-1}\widetilde{w}_{k-1}^T] \approx \mathbb{E}[L_k \mu_k \mu_k^T L_k^T].$$

Not a proof: Motivating form of $\boldsymbol{\Sigma}_{\tilde{v}}$

- Again using EKF as a basis, consider Step 2a:

$$L_k = \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1},$$

  where $\Sigma_{\tilde{z}} = \widehat{C}_k \Sigma_{\tilde{x},k}^- \widehat{C}_k^T + \Sigma_{\tilde{v}}$ is the covariance of the innovation μ_k.
- So, we have that $\Sigma_{\tilde{v}} = \mathbb{E}[\mu_k \mu_k^T] - \widehat{C}_k \Sigma_{\tilde{x},k}^- \widehat{C}_k^T$.
  - Some adaptive EKFs use this relationship.
  - However, the subtraction can lead to non-positive-definite $\Sigma_{\tilde{v}}$ in practice.
- A more robust solution defines the residual: $r_k = z_k - h(\hat{x}_k^+, u_k, \bar{v}_k)$.
  - Note that the residual uses $\hat{x}_k^+$ and the innovation uses $\hat{x}_k^-$.
  - Then, it is possible to derive $\Sigma_{\tilde{v}} = \mathbb{E}[r_k r_k^T + \widehat{C}_k \Sigma_{\tilde{x},k}^+ \widehat{C}_k^T]$.

Smoothing the estimates of $\boldsymbol{\Sigma}_{\widetilde{w}}$ and $\boldsymbol{\Sigma}_{\tilde{v}}$

- Instantaneous estimates of $\Sigma_{\widetilde{w},k}$ and $\Sigma_{\tilde{v},k}$ using these equations tend to be very noisy and introduce instability into the adaptive EKF (AEKF) solution.
- Authors use ad-hoc methods to smooth estimates of $\Sigma_{\widetilde{w},k}$ and $\Sigma_{\tilde{v},k}$ either by averaging over a buffer of $\Sigma_{(\cdot),k}$ values or by using a simple one-pole filter:

$$\Sigma_{(\cdot),k} = \alpha \Sigma_{(\cdot),k-1} + (1-\alpha)\left[\text{New value of } (\cdot)_k\right].$$

- I find that smoothing is very important to getting AEKF to work.
- My experience is that the method to estimate $\Sigma_{\tilde{v}}$ tends to work quite well.
- But I often have trouble with the method to estimate $\Sigma_{\widetilde{w}}$.
  - I think that this is because $\hat{x}_k^+ - \hat{x}_k^-$ is not exactly equal to $\widetilde{w}_{k-1}$.
- The next lesson will introduce AEKF code and share an example: You will see how I modified one step to make it work well for that example.

Summary

- The AEKF seeks to make estimates $\Sigma_{\widetilde{w},k}$ and $\Sigma_{\tilde{v},k}$ of the process-noise and sensor-noise covariance matrices.
- My own experiences with the AEKF have been mixed: sometimes I get good results and other times the filter becomes unstable.
- But, AEKF is very popular in the literature and it is a compelling approach to finding $\Sigma_{\widetilde{w}}$ and $\Sigma_{\tilde{v}}$ if it works well for your nonlinear system.
- It is not difficult to implement, as you will learn in the next lesson, and so it is worth trying.